

ORIGINAL RESEARCH

Socioeconomic Correlates of Incident Heart Failure and Consequent Mortality

The All of Us Program

Yakubu Bene-Alhasan, MD, MPH,^{a,b} Patience Saaka, MD, MPH,^c Isaac Acquah, MD, MPH,^d Sahithi R. Kalluri, MD,^e Samuel Mensah, MD, MPH,^e Ahmad Alkhatib, MD,^e Lida Koskina, MD,^e Leen Othman, MD,^e Albert D. Osei, MD, MPH,^f Christie M. Ballantyne, MD,^a Vijay Nambi, MD, PhD,^a Khurram Nasir, MD, MPH, MSc,^g Kenneth J. Mukamal, MD, MPH^b

ABSTRACT

BACKGROUND Although socioeconomic status (SES) is a known determinant of cardiovascular disease, the independent risks of incident heart failure (HF) and consequent mortality portended by individual and neighborhood measures of SES remain less established.

OBJECTIVES This study sought to evaluate the prospective associations of SES with HF in adults without HF and SES with all-cause mortality in those with HF.

METHODS The authors identified adults from the National Institutes of Health-run All of Us Research Program (2018-present) who consented to share their electronic health records. Among 280,431 participants free of HF and 10,550 participants with prevalent HF, Cox proportional hazards models assessed associations of income, education, and Area Deprivation Index (ADI) with risks of incident HF and mortality.

RESULTS Over 41 ± 23 months, 6,783 of 280,431 participants developed HF. Compared with household income $\geq \$200,000$, risk was higher for \$100,000 to $< \$200,000$ (HR: 1.29; 95% CI: 1.12-1.48), \$50,000 to $< \$100,000$ (HR: 1.82; 95% CI: 1.60-2.09), \$25,000 to $< \$50,000$ (HR: 2.24; 95% CI: 1.95-2.57), and $< \$25,000$ (HR: 3.02; 95% CI: 2.63-3.47). Relative to college graduates, risk was higher for those with some college (HR: 1.37; 95% CI: 1.28-1.46), General Educational Development (HR: 1.46; 95% CI: 1.36-1.58), and less than high school (HR: 1.40; 95% CI: 1.26-1.55). Compared with the least deprived tertile of ADI, risk was higher in the middle tertile (HR: 1.21; 95% CI: 1.14-1.29) and the most deprived tertile (HR: 1.18; 95% CI: 1.11-1.26). Non-White participants and residents of the most deprived ADI tertiles experienced less benefit from higher education. Among those with prevalent HF, each \$10,000 increase in income was associated with 3% lower all-cause mortality.

CONCLUSIONS Lower income, education, and high neighborhood deprivation independently associate with incident HF, whereas only income was associated with mortality. Higher education was less protective in non-White participants and individuals residing in deprived neighborhoods. Addressing these disparities is essential to reducing HF burden and consequent mortality. (JACC Heart Fail. 2026;■:103012) © 2026 the American College of Cardiology Foundation. Published by Elsevier. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

From the ^aSection of Preventive Cardiology, Baylor College of Medicine, Houston, Texas, USA; ^bDepartment of Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts, USA; ^cDepartment of Global Health and Population, Harvard T.H. Chan School of Public Health, Boston, Massachusetts, USA; ^dDivision of Cardiology, University of Colorado, Aurora, Colorado, USA; ^eDepartment of Internal Medicine, MedStar Health Georgetown University, Baltimore, Maryland, USA; ^fDivision of Cardiology, University of Pittsburgh, Pittsburgh, Pennsylvania, USA; and the ^gDivision of Cardiology, Houston Methodist, Houston, Texas, USA.

**ABBREVIATIONS
AND ACRONYMS****ADI** = Area Deprivation Index**BMI** = body mass index**CVD** = cardiovascular disease**EHR** = electronic health record**GED** = General Educational
Development**HEA** = highest educational
attainment**HF** = heart failure**SES** = socioeconomic status**THI** = total household income

Hear failure (HF) remains a leading cause of morbidity and mortality globally. According to the HFSA (Heart Failure Society of America), ~6.7 million Americans >20 years of age had HF in 2020, and this number is expected to rise to 8.7 million by 2030, representing a ~30% increase.¹ Globally, the prevalence of HF increased by 29% between 2010 and 2019, and North America remains the region with the highest age-adjusted prevalence.² Hence, defining predictors of HF remains a compelling public health priority.

Socioeconomic status (SES) is typically measured at the individual level by correlates such as educational attainment and household income, and at the neighborhood level by indices like the widely used Area Deprivation Index (ADI).^{3,4} Both individual- and neighborhood-level measures of SES have emerged as determinants of cardiovascular disease (CVD) outcomes, including HF, and have been shown to be predictors of CVD outcomes in the United States and globally.^{2,5,6} However, prior studies have provided cross-sectional evidence of an association, or included individual or neighborhood measures, but not both concurrently. Moreover, the existing studies have followed cohorts established decades ago; given the substantial sociopolitical and economic changes that have since occurred, their findings may have limited relevance today. Indeed, these changes in individual and neighborhood SES warrant renewed investigation.

Leveraging data from the NIH (National Institutes of Health)-run All of Us Research Program, we evaluated the association of individual SES, defined by highest educational attainment (HEA) and total household income (THI), and neighborhood SES, defined by ADI, with incident HF and, in those with prevalent HF, all-cause mortality. By examining this large, diverse cohort, we sought to elucidate the extent and nuance through which SES influences HF incidence and consequent mortality, thereby informing targeted public health, clinical, and policy interventions aimed at reducing cardiovascular health disparities.

METHODS

STUDY POPULATION. The All of Us Research Program is an NIH-run population-based open cohort of

community-dwelling adults⁷ launched in 2018 as part of the NIH's Precision Medicine Initiative, and focuses on groups historically underrepresented in biomedical research. Inclusion criteria are ≥18 years of age, residence in the U.S. or U.S. territories, and the capacity to provide informed consent. Incarcerated individuals at the time of enrollment are excluded. Enrollment began in 2018 and is ongoing. Participants complete a baseline demographic survey, complete baseline physical measurement, provide biospecimens, and are later invited to complete additional surveys about lifestyle and social determinants of health. At the time of enrollment, they provide consent to share their electronic health record (EHR) and can choose to rescind this decision at any point in time. The research program assessed health care claims at the time of entry and has continued to update those. All such claims include dates of service, enabling analysis of both diagnoses that were prevalent at the time of study entry and of those that were incident after enrollment. We used version 8 of the Curated Data Repository, which includes participant information from the time of enrollment to October 1, 2023.

The Curated Data Repository contained data on 627,584 participants at the time of our analyses (August 1, 2025). Of these, 393,596 participants consented to the release of their EHR, and 377,171 did not have HF at baseline. We excluded an additional 96,740 participants (25.6%) for missing data on any covariate in our main model, resulting in a final population of 280,431 participants; however, we retained these individuals in sensitivity analyses using multiple imputation. The All of Us Institutional Review Board reviewed and approved the All of Us protocol and materials. Our study was exempted from local Institutional Review Board review because it did not directly involve participants, only their deidentified data. All study procedures were performed in accordance with the ethical standards set by the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

EXPOSURE. Individual SES data were collected via enrollment surveys with questions about the range of their THI and HEA. THI was categorized as follows: <\$25,000, \$25,000 to <\$50,000, \$50,000 to <\$100,000, \$100,000 to <\$200,000, and ≥\$200,000. THI of ≥\$200,000 was used as the

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reference group. HEA was categorized as less than General Educational Development (GED), GED, college years 1 to 3, and college degree. Those with a college degree were the reference group.

Neighborhood SES was measured by the ADI—a composite measure of communities' socioeconomic deprivation. The ADI is a validated measure of neighborhoods' SES and has been extensively used.^{8,9} In All of Us, the index was created using principal component analysis of 6 different American Community Survey measures that included the neighborhood's median income, fraction of vacant housing, fraction of households living below poverty level, fraction with no health insurance, fraction on assisted income, and fraction of those >25 years of age to have graduated high school.⁸ The first component explained over 60% of the total variance present and was rescaled to range from 0 to 1. The ADI was obtained from this ratio with a higher ratio representing higher deprivation. Participants' residential zip code was reduced to the first 3 digits to preserve their privacy. These zip code data were mapped to their computed ADI. We categorized the ADI into tertiles. The lowest tertile (least deprived zip codes) was the reference group.

OUTCOME. The outcome variable in the primary analysis was a new EHR diagnosis of HF during follow-up. We required HF to be documented (using the International Classification of Diseases [ICD] I50) on 2 outpatient visits or a single inpatient visit to define HF, as used elsewhere.¹⁰ In our secondary analyses among individuals with prevalent HF, the outcome variable was all-cause mortality, obtained from the EHR and ascertained by the All of Us Research Program.

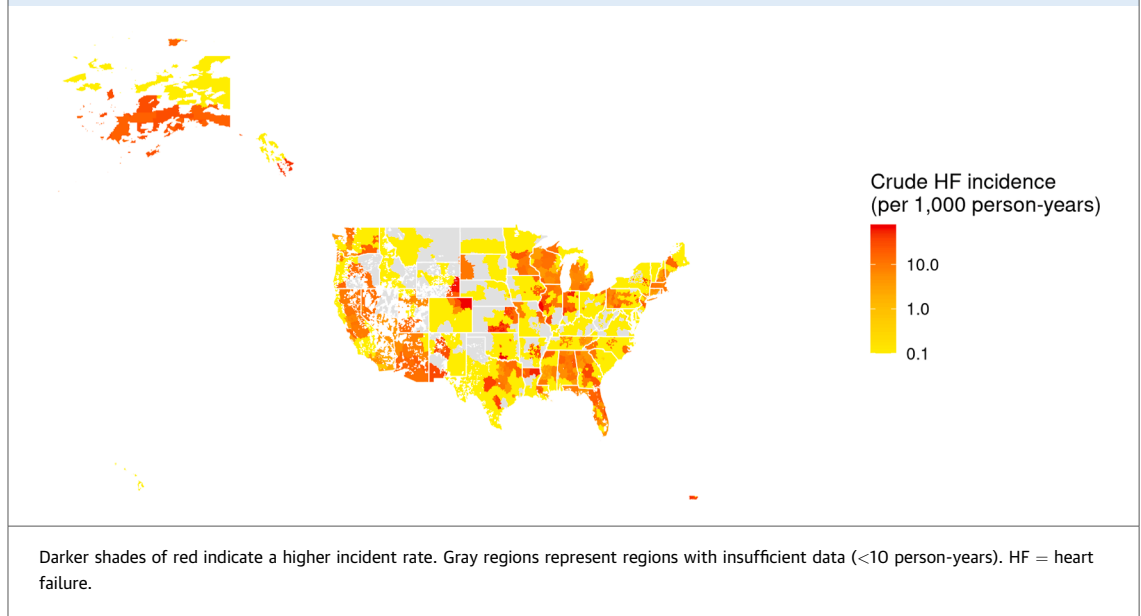
COVARIATES. Sociodemographic and lifestyle information were obtained from the baseline questionnaires at enrollment. The research program computed body mass index (BMI) from weight and height measurements. Information on alcohol consumption was obtained by the AUDIT-C (Alcohol Use Disorders Identification Test-Concise) instrument.¹¹ Information on cigarette smoking history was collected by the lifestyle survey. Prevalent medical conditions including diabetes mellitus (ICD E10 and E11), hypertension (ICD I10), and dyslipidemia (ICD E78) were defined as EHR diagnosis of these conditions, because those records were defined as EHR diagnoses of these conditions from when those records became available to baseline. Similarly, 2 EHR documentations of these conditions or a single inpatient documentation was used to define their presence.

STATISTICAL ANALYSES. We described individual and neighborhood SES distribution within strata, by demography, by substance use, and by prevalent cardiovascular risk factors, and provided summary statistics including means with SDs (if continuous) or number with proportions (if categorical). We also showed crude incident rates of HF by zip code within the United States (Figure 1).

We built sequential multivariable Cox proportional hazards models to investigate the association between SES and incident HF. Time to event or censoring was calculated as the time from completing the survey on SES to the date of incident HF, date of death, or end of follow-up (October 1, 2023). The proportional hazards assumption was tested graphically using Schoenfeld residuals. We tested for nonlinearity of variables using restricted cubic splines and categorized ADI into tertiles and HEA into easily comprehended categories based on our findings of nonlinearity. We also provided results of these modeled linearly. Our models included plausibly confounding covariates based on the known association in published reports. The primary model included the following: ADI, THI, and HEA, all in the same model, and age, sex, race, Hispanic ethnicity, and household size. Our secondary models added cigarette smoking status and alcohol consumption to the primary model; and the tertiary model added hypertension, diabetes mellitus, and BMI (modeled as linear and quadratic terms) to the secondary model.

To test the sensitivity of our findings to missing data, we repeated our primary analyses using multiple imputation by chained equations. The imputation model included all variables in the primary model and the outcome variable (time to HF). We generated 5 imputed data sets using predictive mean matching for continuous variables, logistic regression for binary variables, and polytomous regression for categorical variables. Our primary model was fitted within each imputed data set, and estimates were combined using Rubin's rules to obtain pooled HRs with 95% CIs.

In subgroup analyses, we tested for interaction between SES measures and sex, and race, by including product terms to our primary model, and evaluated the association between SES and HF in subgroups of sex (male and female) and race (Black, White, and other [comprising Middle Eastern and North African, Asian, Native Hawaiian and other Pacific Islanders, and other racial groups combined due to low event rates in individual groups]) that showed evidence of interaction. In other subgroup analyses, we evaluated the joint associations between the individual SES measures and HF within tertiles of ADI.

FIGURE 1 Log-Transformed Crude Incident HF Rate Across the United States by Zip Code

In a complementary analysis, we assessed all-cause mortality among those with prevalent HF at baseline, while adjusting for the same variables in our primary model. Time to event for this analysis was defined as time from baseline to death or end of follow-up. In a post hoc analysis, also among those with prevalent HF, we evaluated the potential bidirectional association between SES and HF² using an ordinal logistic regression model with income as the outcome and time (in years) from HF diagnosis to SES assessment as the exposure, and adjusted for covariates similar to those in our primary model (age, race, ethnicity, sex, and educational attainment).

All analyses were carried out in R (R Core Team, 2021). Statistical significance was defined as 2-tailed value of $P \leq 0.05$ for main analyses, and Bonferroni-corrected limits were used for interactions.

RESULTS

Among the 280,431 participants in our primary analytical sample of incident HF, approximately one-third were male, 16% were Black, 63% were White, 3% were Asian, and 15% were Hispanic. The mean age of participants was 52 ± 17 years. Educational attainment was generally high, with more than one-half holding a college degree, 27% reporting 1 to 3 years of college, 16% reporting receiving a GED, and 6% having less than a GED. Household income showed wide variation: 28% reported earning <\$25,000 annually, whereas 9% earned at

least \$200,000. The mean ADI was 0.32 ± 0.06 , with values ranging from 0.15 to 0.63 ([Supplemental Table 1](#)).

Compared with included participants, those who were excluded differed in several important respects. Excluded individuals were more often Hispanic (29% vs 15%), less likely to be White (31% vs 63%), and less likely to hold a college degree (28% vs 51%). They were also more likely to have less than a GED (17% vs 6%). Household income was generally lower among excluded individuals ([Supplemental Table 1](#)).

Clear socioeconomic gradients were evident when comparing participants by income ([Table 1](#)). Those with higher income were generally older, thinner, more likely to be White, and more likely to be covered by private insurance ([Table 1](#)).

Similar socioeconomic patterns were observed when stratifying by education. Participants with less than a high school education had higher prevalence of diabetes and hypertension, a higher mean BMI, and were more likely to be covered by public insurance than those holding a college degree ([Supplemental Table 2](#)).

Over a mean follow-up of 41 ± 23 months, 6,783 participants developed incident HF. In [Figure 1](#), we show a graphical description of the crude incidence of HF by geographic area. Higher rates are seen in participants enrolled from the Midwest, Southeast, and Southwest regions.

INDIVIDUAL MEASURES AND HF. THI had an inverse dose-response relationship with incident HF in our

TABLE 1 Baseline Characteristics by Total Household Income

	≥\$200,000 (n = 24,932)	\$100,000-\$200,000 (n = 56,344)	\$50,000-\$100,000 (n = 69,008)	\$25,000-\$50,000 (n = 52,304)	<\$25,000 (n = 77,843)	P Value
Male	11,525 (46.2)	22,679 (40.3)	25,035 (36.3)	17,385 (33.2)	29,543 (38.0)	<0.001
Age at baseline, y	54.38 ± 14.37	54.21 ± 15.73	53.88 ± 17.31	50.29 ± 18.41	48.07 ± 16.06	<0.001
Race						<0.001
Asian	1,530 (6.1)	2,510 (4.5)	2,366 (3.4)	1,414 (2.7)	1,264 (1.6)	
Black	610 (2.4)	2,491 (4.4)	5,970 (8.7)	8,670 (16.6)	27,391 (35.2)	
Middle Eastern North African	214 (0.9)	363 (0.6)	380 (0.6)	328 (0.6)	353 (0.5)	
Missing	1,023 (4.1)	3,329 (5.9)	6,898 (10.0)	9,557 (18.3)	17,169 (22.1)	
Other ^a	757 (3.0)	1,540 (2.7)	2,081 (3.0)	1,681 (3.2)	2,664 (3.4)	
White	20,798 (83.4)	46,111 (81.8)	51,313 (74.4)	30,654 (58.6)	29,002 (37.3)	
Ethnicity						<0.001
Hispanic	1,246 (5.0)	3,989 (7.1)	7,751 (11.2)	10,332 (19.8)	17,928 (23.0)	
Missing	259 (1.0)	558 (1.0)	885 (1.3)	737 (1.4)	1,267 (1.6)	
Non-Hispanic	23,427 (94.0)	51,797 (91.9)	60,372 (87.5)	41,235 (78.8)	58,648 (75.3)	
Household size	1.88 ± 1.25	1.73 ± 1.28	1.58 ± 1.38	1.73 ± 1.63	1.70 ± 1.75	<0.001
Incident heart failure	257 (1.0)	779 (1.4)	1,526 (2.2)	1,416 (2.7)	2,805 (3.6)	<0.001
Health insurance						<0.001
Employer or union	16,706 (74.3)	34,547 (67.7)	33,958 (55.3)	16,287 (36.8)	5,084 (8.3)	
Medicaid	105 (0.5)	585 (1.1)	3,123 (5.1)	9,269 (21.0)	35,848 (58.5)	
Medicare	3,249 (14.5)	9,586 (18.8)	14,674 (23.9)	10,604 (24.0)	11,879 (19.4)	
None	5 (0.0)	17 (0.0)	62 (0.1)	202 (0.5)	784 (1.3)	
Other	857 (3.8)	2,931 (5.7)	4,689 (7.6)	4,155 (9.4)	4,991 (8.1)	
Purchased	1,555 (6.9)	3,339 (6.5)	4,899 (8.0)	3,698 (8.4)	2,715 (4.4)	
AUDIT-C	3.16 ± 1.78	2.90 ± 1.83	2.71 ± 1.91	2.62 ± 2.11	2.66 ± 2.57	<0.001
Smoking status						<0.001
Never	17,858 (73.8)	37,445 (68.6)	42,530 (63.7)	30,230 (59.8)	35,666 (47.7)	
Former	3,741 (15.5)	10,281 (18.8)	14,531 (21.7)	12,221 (24.2)	23,182 (31.0)	
Current	2,606 (10.8)	6,835 (12.5)	9,751 (14.6)	8,104 (16.0)	15,964 (21.3)	
Diabetes mellitus	1,273 (5.1)	4,163 (7.4)	7,070 (10.2)	6,477 (12.4)	11,052 (14.2)	<0.001
Hypertension	4,920 (19.7)	12,924 (22.9)	18,069 (26.2)	13,981 (26.7)	20,694 (26.6)	<0.001
Body mass index	26.98 ± 5.51	28.59 ± 6.45	29.84 ± 7.19	30.74 ± 7.91	30.84 ± 8.53	<0.001
Follow-up time, wk	150.40 ± 96.25	150.00 ± 95.37	154.86 ± 94.93	165.01 ± 94.08	187.01 ± 85.71	<0.001
Highest educational attainment						<0.001
College graduate and beyond	22,637 (90.8)	44,565 (79.1)	42,422 (61.5)	20,157 (38.5)	12,846 (16.5)	
College years 1-3	1,874 (7.5)	9,432 (16.7)	19,353 (28.0)	19,406 (37.1)	24,161 (31.0)	
12th grade or GED	382 (1.5)	2,210 (3.9)	6,591 (9.6)	10,388 (19.9)	26,369 (33.9)	
Less than high school	39 (0.2)	137 (0.2)	642 (0.9)	2,353 (4.5)	14,467 (18.6)	
ADI	0.29 ± 0.07	0.30 ± 0.06	0.31 ± 0.06	0.32 ± 0.06	0.34 ± 0.06	<0.001

Values are mean ± SD or n (%), unless otherwise indicated. ^aOthers include Native Hawaiian or other Pacific Islanders, other racial groups, and >1 racial group.

ADI = Area Deprivation Index; AUDIT-C = Alcohol Use Disorders Identification Test-Concise score; GED = General Educational Development.

main model (Table 2). We found similarly significant but attenuated risks that maintained their dose-response associations in the secondary and tertiary models. Our spline analyses showed an approximately linear association between THI and risk of HF, and each \$10,000 increment in household income was associated with a 5% lower risk of incident HF (Table 2). Participants with less than a GED, those with a GED, and those who had attended 1 to 3 years of college all had 37% to 46% higher risk of HF than college graduates; we did not observe a significant difference among these groups.

ADI AND HF. Residing in more deprived regions was associated with a higher risk of HF. This was, however, not a dose-response association. Compared with those residing in areas in the lower tertile of ADI (less deprived neighborhoods), those in the middle and upper tertiles had a 21% and 18% higher risk of incident HF (Table 2), respectively. These higher risks persisted even after further adjusting for substance use and prevalent CVD risk factors.

ANALYSES AMONG INDIVIDUALS WITH PREVALENT HF. In our complementary analysis among those with prevalent HF, we found an inversely linear

TABLE 2 Models Showing Adjusted HRs of Incident HF by Strata of Total Household Income, Highest Educational Attainment, and Deprivation Index

	Primary Model (Main Model) (6,783 HF Events) (n = 280,431)		Secondary Model (4,677 HF Events) (n = 220,173)		Tertiary Model (4,418 HF Events) (n = 207,967)	
	HR (95% CI)	P Value	HR (95% CI)	P Value	HR (95% CI)	P Value
Adjusted HRs of HF by strata of total household income						
Total household income modeled linearly (per \$10,000 higher)	0.95 (0.95-0.96)	<0.001 ^a	0.96 (0.95-0.96)	<0.001 ^a	0.97 (0.96-0.97)	<0.001 ^a
>\$200,000	Ref.		Ref.		Ref.	
\$100,000-\$200,000	1.29 (1.12-1.48)	<0.001	1.27 (1.08-1.48)	0.003	1.17 (1.00-1.37)	0.06
\$50,000-\$100,000	1.82 (1.60-2.09)	<0.001	1.76 (1.51-2.04)	<0.001	1.51 (1.29-1.76)	<0.001
\$25,000-\$50,000	2.24 (1.95-2.57)	<0.001	2.07 (1.77-2.42)	<0.001	1.70 (1.45-2.00)	<0.001
<\$25,000	3.02 (2.63-3.47)	<0.001	2.81 (2.40-3.30)	<0.001	2.36 (2.01-2.78)	<0.001
Adjusted HRs of HF by strata of highest educational attainment						
Highest educational attainment modeled linearly (per grade higher)	0.95 (0.94-0.96)	<0.001 ^a	0.95 (0.94-0.96)	<0.001 ^a	0.97 (0.96-0.98)	<0.001 ^a
College graduate and beyond	Ref.		Ref.		Ref.	
College	1.37 (1.28-1.46)	<0.001	1.32 (1.23-1.43)	<0.001	1.15 (1.06-1.25)	<0.001
12th grade or GED	1.46 (1.36-1.58)	<0.001	1.40 (1.28-1.54)	<0.001	1.25 (1.14-1.37)	<0.001
Less than high school	1.40 (1.26-1.55)	<0.001	1.34 (1.18-1.52)	<0.001	1.17 (1.03-1.34)	0.02
Adjusted HRs of HF by ADI						
ADI, modeled linearly (per SD higher)	1.09 (1.06-1.12)	<0.001	1.07 (1.04-1.11)	<0.001	1.09 (1.05-1.12)	<0.001
Lower tertile	Ref.		Ref.		Ref.	
Middle tertile	1.21 (1.14-1.29)	<0.001	1.20 (1.12-1.29)	<0.001	1.19 (1.10-1.28)	<0.001
Upper tertile	1.18 (1.11-1.26)	<0.001	1.15 (1.06-1.24)	0.006	1.23 (1.13-1.33)	<0.001

Primary model includes highest educational attainment, total household income, and ADI, and adjusts for age, sex, race, Hispanic ethnicity, and household size. Secondary model includes variables in the primary model + smoking status (never, former, current) and AUDIT-C scores. Tertiary model includes variables in the secondary model + hypertension, diabetes mellitus, and body mass index. ^aP value of linear trends.

HF = heart failure; Ref. = Reference; other abbreviations as in Table 1.

TABLE 3 Model Showing Adjusted HRs of All-Cause Mortality Among Those With Prevalent Disease at Baseline According to Total Household Income, Highest Educational Attainment, and Deprivation Index

	Primary Model (Main Model) (505 Deaths) (n = 10,550)	
	HR (95% CI)	P Value
Total household income modeled linearly (per \$10,000 higher)	0.97 (0.95-0.99)	0.003
>\$200,000	Ref.	Ref.
\$100,000-\$200,000	1.74 (0.89-3.42)	0.11
\$50,000-\$100,000	1.64 (0.85-3.16)	0.14
\$25,000-\$50,000	1.92 (0.99-3.71)	0.052
<\$25,000	2.40 (1.24-4.64)	0.009
Highest educational attainment modeled linearly (per grade higher)	0.98 (0.94-1.01)	0.22
College graduate and beyond	Ref.	Ref.
College	1.09 (0.86-1.38)	0.48
12th grade or GED	1.29 (1.00-1.66)	0.049
Less than high school	0.94 (0.64-1.38)	0.74
ADI, modeled linearly (per SD higher)	1.03 (0.94-1.14)	0.48
Lower tertile	Ref.	Ref.
Middle tertile	0.82 (0.65-1.02)	0.08
Upper tertile	1.08 (0.86-1.34)	0.52

Model includes highest educational attainment, total household income, and ADI, and adjusts for age, sex, race, Hispanic ethnicity, and household size.

Abbreviations as in Tables 1 and 2.

relationship between THI and all-cause mortality. Those making <\$25,000 had a 140% higher risk of all-cause mortality than those making ≥\$200,000. Each \$10,000 higher THI was associated with a 3% lower risk of mortality. In contrast, the relationships with HEA and ADI were not graded nor statistically significant (Table 3).

In the post hoc ordinal regression analysis, the longer in the past that a HF diagnosis was first identified, the higher the odds of being in a lower-income category (OR per year: 1.11; 95% CI: 1.09-1.13).

SENSITIVITY ANALYSIS. Analyses across the 5 imputed data sets that adjusted for the same covariates in our main model (model 1) yielded consistent results to complete-case analyses, and pooled estimates are presented in Table 4. For each stratum compared with the reference, the associations were attenuated for THI, but stronger for HEA and ADI.

SUBGROUP ANALYSES. In our subgroup analyses, the Bonferroni-corrected limits of the models were 0.017. There was evidence of interaction of all SES measures by race (all $P < 0.001$). However, we found evidence of interaction by sex and THI ($P < 0.001$) and ADI tertiles ($P = 0.002$), but not HEA ($P = 0.08$). Females had an inverse dose-response association

between THI and tertiles of ADI, and incident HF. For each level of THI compared with the referent, the HR of HF risk was higher among females compared with males (Supplemental Figure 1). Whites exhibited an inverse dose-response risk of HF with THI, HEA, and ADI tertiles. Blacks and other racial groups also exhibited an inverse dose-response association between THI and HF, but not HEA (Supplemental Figure 2).

In our joint analyses, we found evidence of interaction by tertiles of ADI and HEA ($P = 0.02$) but not THI ($P = 0.04$, Bonferroni limit: $P < 0.025$). HEA was associated with an inverse dose-response risk of HF only in those residing in the lower tertile (Supplemental Figure 3).

DISCUSSION

We found strong associations between SES and HF in a contemporary U.S. cohort. Lower income, lower educational attainment, and higher neighborhood deprivation were consistently linked to higher HF incidence, with additional mortality risk among those with prevalent HF and low income (Central Illustration). These findings emphasize the importance of SES as a determinant in HF pathogenesis and outcomes, and highlight potential differences across race, sex, and neighborhood context.

HF often represents the final stage of CVD, and prior research has shown that low SES is associated with higher prevalence of diabetes, hypertension, dyslipidemia, and coronary artery disease, all on the pathway to HF.^{12,13} Although studies directly examining SES and HF are limited, our findings align with existing evidence. For example, Lawson et al¹⁴ reported that in newly diagnosed HF patients, the most deprived groups had more comorbidities than the most affluent. Domengé et al¹⁵ found income and education to be the strongest predictors of HF in a large French cohort. Studies using neighborhood-level SES measures have also demonstrated similar associations with HF incidence.¹⁶

THI AND HF. Abdalla et al,¹⁷ using nationally representative U.S. data, reported 10% lower odds of HF in the highest vs lowest income group from 1999 to 2016. We found a similar dose-response relationship: each \$10,000 increase in household income was associated with a 5% lower risk of incident HF and a 3% lower risk of mortality in those with prevalent HF. THI was the strongest and most consistent SES predictor of HF. This aligns with prior findings: Lewis et al¹⁸ reported low income was associated with coronary heart disease regardless of education, and Mathews et al⁶ found income strongly predicted

TABLE 4 Table of Pooled HRs From Multiple Imputation Analysis of Main Model (Model 1)

	Main Model (Model 1)	
	HR (95% CI)	P Value
Total household income		
>\$200,000	Ref.	Ref.
\$100,000-\$200,000	0.95 (0.83-1.08)	0.42
\$50,000-\$100,000	1.23 (1.08-1.40)	0.003
\$25,000-\$50,000	1.37 (1.23-1.54)	<0.001
<\$25,000	1.69 (1.53-1.88)	<0.001
Highest educational attainment		
College graduate and beyond	Ref.	Ref.
College	1.53 (1.45-1.62)	<0.001
12th grade or GED	1.61 (1.52-1.71)	<0.001
Less than high school	1.62 (1.49-1.76)	<0.001
ADI		
Lower tertile	Ref.	Ref.
Middle tertile	1.28 (1.21-1.34)	<0.001
Upper tertile	1.23 (1.17-1.30)	<0.001

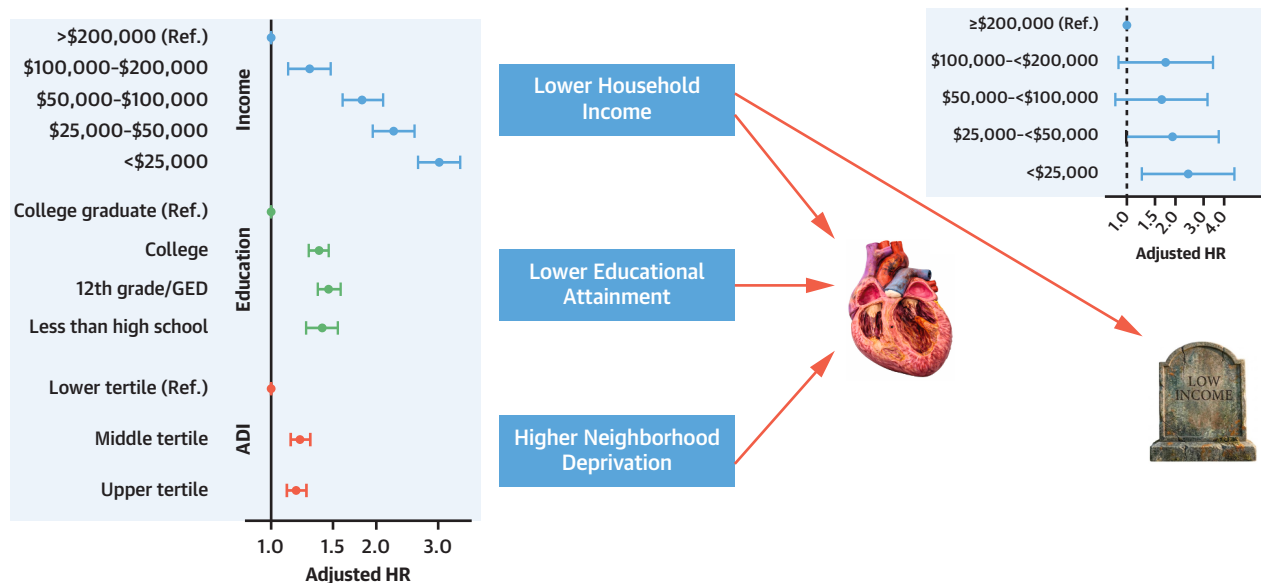
Model includes highest educational attainment, total household income, and ADI, and adjusts for age, sex, race, Hispanic ethnicity, and household size.
Abbreviations as in Tables 1 and 2.

mortality in HF, whereas education and ADI were weaker predictors. Although overall U.S. household income has grown, nearly one-half of aggregate income is now concentrated in the upper class, up from one-third in 1970.¹⁹ This widening inequality risks worsening SES-driven health disparities by limiting economic mobility, educational opportunities, and access to quality health care.

Income reflects access to resources, health care, and living conditions, and is closely tied to insurance status.²⁰ In our cohort, the lowest income group was most often uninsured (Table 1). Insurance status was not modeled due to collinearity with income and education. Nonetheless, as shown in Table 1 and Supplemental Table 2, lower SES groups disproportionately relied on Medicare and Medicaid, whereas higher SES groups more often had employer- or union-sponsored coverage, highlighting that the most vulnerable populations, who already face higher HF risk and worse outcomes, depend heavily on public insurance. Policies restricting Medicare or Medicaid would likely deepen existing disparities and raise health care costs. Financial barriers and lack of insurance can restrict care, reduce adherence to therapies for HF risk factors, and worsen post-myocardial infarction outcomes,²¹⁻²⁵ whereas chronic financial stress may further contribute to adverse cardiovascular remodeling.²⁶ This relationship between income and SES is also evident at the global level, where a graded increase in the prevalence of HF is observed across national income quartiles.²

CENTRAL ILLUSTRATION Lower Socioeconomic Status Increases Risk of Incident Heart Failure and Subsequent Mortality

Risk of incident heart failure in 280,431 adults without heart failure | Risk of all-cause mortality in 10,550 adults with heart failure



Lower income, education, and higher neighborhood deprivation independently predict incident heart failure. Protective effects of higher education were attenuated in non-White participants and residents of more deprived neighborhoods.

Only low income predicts mortality in those with heart failure.

Bene-Alhasan Y, et al. JACC Heart Fail. 2026;■(■):103012.

In those without heart failure, participants with lower total household income, lower educational attainment, and higher neighborhood deprivation had a higher risk of incident heart failure. This risk varied by race and neighborhood deprivation. In those with prevalent heart failure, lower income was associated with a higher risk of all-cause mortality. ADI = Area Deprivation Index; GED = General Educational Development; Ref. = Reference.

EDUCATIONAL ATTAINMENT AND HF. We found strong independent associations between lower HEA and incident HF, even after adjusting for income and other confounders. Education influences health through multiple pathways, including employment, income, and health literacy.^{27,28} Lower HEA is often linked to reduced health literacy, which may limit comprehension of health messages, preventive behaviors, and therapy adherence. Conversely, better health literacy promotes healthier lifestyles and lowers cardiovascular risk. Health education should be tailored to populations with limited health literacy. Importantly, declining college graduation rates in the United States²⁹ may further widen disparities in HF incidence if unaddressed.

Subgroup analyses revealed differential associations across populations. White participants and

those in less deprived areas showed an inverse dose-response relationship between HEA and HF incidence. In contrast, among Black participants and other racial groups, those with a GED had higher risk than those with less than a GED. Prior studies have shown similar patterns. Azap et al³⁰ reported no link between education and ideal cardiovascular health, a composite of several CVD risk factors, in Black males, whereas Johnson et al³¹ found education was associated with better cardiovascular health in White participants but not Black participants. These findings support the theory that racial minorities may not benefit from higher SES to the same extent as White participants,³² and that interventions based solely on HEA are unlikely to reduce HF incidence in these groups. A multipronged approach incorporating other SES dimensions is needed.

ADI AND HF. We observed a parabolic relationship between ADI and HF, with the highest risk in the middle tertile and lowest in the least deprived areas. Previous studies also reported higher HF risk in deprived neighborhoods.^{33,34} Because ADI incorporates zip code-level income and education, it reflects neighborhood-level socioeconomic conditions.

Beyond economic measures, neighborhoods influence HF risk through physical and social attributes. Walkability, recreational spaces, and access to healthy foods can reduce CVD risk, whereas social factors like safety, cohesion, and community support also play a role³⁵; high neighborhood safety has been linked to decreasing BMI,³⁶ whereas low social cohesion is associated with greater waist circumference.³⁷ These CVD risk factors likely mediate the ADI-HF relationship; however, the association remained significant after adjusting for multiple CVD risk factors, suggesting additional pathways warrant further study.

Notably, as with education, Black participants in the highest ADI tertile did not experience higher HF risk compared with those in the lowest tertile, further highlighting racial differences in SES benefits. An inverse dose-response relationship between education and HF was only observed in the lowest ADI tertile, suggesting that the protective effect of education may be more pronounced in less deprived neighborhoods.

A BIDIRECTIONAL RELATIONSHIP. We focused on the path from SES to HF and subsequent mortality. Nonetheless, the association is likely bidirectional, as suggested by our post hoc analysis. Low SES may increase the risk of developing HF, and prevalent HF may, in turn, exacerbate socioeconomic disadvantage through increased morbidity and the substantial financial burden borne by patients and their caregivers,² creating a vicious cycle between SES, HF, and mortality.

IMPLICATIONS. Our findings have important implications for public health, policy, and clinical practice. In clinical care, routine SES screening and tailored patient education are essential. Culturally sensitive, community-driven strategies, including outreach by trained community health workers, can improve engagement in underserved groups. Although digital tools like telemedicine may expand access, equitable deployment is critical to avoid worsening disparities.

Potential solutions would not be straightforward, given the nonprotective impact of HEA on HF incidence among racial minorities and individuals residing in deprived neighborhoods. Accordingly,

potential solutions should adopt a multipronged approach that addresses all components of SES. Initiatives that improve income and educational attainment through targeted programs and health literacy efforts, particularly for disadvantaged populations, should be considered. Enhancing access to quality schools and colleges, reducing dropout rates, and expanding financial support, including affordable student loans, can help bridge social gaps, promote economic mobility, and ultimately reduce HF risk.

At the community level, urban planning should prioritize affordable neighborhoods with features that support cardiovascular health, including walkability, recreational spaces, and social cohesion. Geographic “hotspotting” has proven effective³⁸ for identifying high-burden areas for targeted interventions, as reflected by regional clustering of HF incidence in our data, which interestingly mirrors CVD mortality³⁹ (Figure 1). Both highlight the higher risk of HF (and CVD mortality) in the Midwest and Southeast. Clinicians, policymakers, and other stakeholders in these regions should prioritize clinical and public health interventions including education, screening, and treatment of pre-HF and HF risk factors, particularly among individuals with low SES.

STUDY STRENGTHS AND LIMITATIONS. Our study has several strengths. To our knowledge, this is the largest prospective investigation of SES and HF, incorporating both individual and neighborhood measures for a comprehensive assessment. Unlike prior studies that focused on composite cardiovascular endpoints or prevalent HF, we specifically examined incident HF, providing insight into risk development over time. Leveraging the diverse All of Us cohort, which includes substantial representation of racial and ethnic minorities, enhances generalizability and highlights disparities across sociodemographic groups. The surveys were recently administered, providing contemporary, granular data on income and education relevant to current public health contexts.

The study also has limitations. Education and income were self-reported, which may introduce misclassification, and EHR-based outcome data are subject to measurement bias; however, we mitigated this by requiring 2 outpatient or 1 inpatient encounter to define HF. Baseline SES measures were used, so temporal changes were not captured; however, SES is generally stable over time.⁴⁰ Approximately one-quarter of participants had missing data, but multiple imputation produced consistent results

with complete-case analyses. Participants with missing SES data were more likely to have lower education and income, live in deprived neighborhoods, and have higher prevalence of cardiovascular risk factors, highlighting potential biases. Finally, despite the prospective design and observed dose-response relationships, residual confounding cannot be excluded, limiting causal inference.

Although we evaluated for a potentially bidirectional relationship, this was limited by the absence of longitudinal SES measurements. Future studies incorporating longitudinal assessments of socioeconomic measures are essential to characterize these dynamic associations more fully.

CONCLUSIONS

This study provides compelling evidence that socioeconomic disparities in income, education, and neighborhood deprivation independently impact HF incidence significantly, whereas lower income predicts subsequent mortality. These findings support the critical need for multidimensional strategies to mitigate socioeconomic barriers in cardiovascular health. Future efforts should integrate socioeconomic screenings into routine clinical assessments, enhance community engagement through tailored interventions, and implement policies that supplement incomes, make higher education more accessible and financially feasible, and provide low-income housing in healthy environments for low SES populations. Addressing these determinants holistically can not only reduce HF incidence, but also promote health equity and ultimately improve cardiovascular outcomes at the population level.

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ADDRESS FOR CORRESPONDENCE: Dr Yakubu Bene-Alhasan, Center for Cardiometabolic Disease Prevention, Baylor College of Medicine, 6655 Travis Street, Suite 320, Houston, Texas 77030, USA. E-mail: Yakubu.bene-alhasan@bcm.edu. X handle: [@yakoubenehasan](https://twitter.com/yakoubenehasan).

PERSPECTIVES

COMPETENCY IN PATIENT CARE AND

PROCEDURAL SKILLS: Data on SES at the individual level, using household income and educational attainment, and neighborhood level using the ADI, derived from zip code data, should be routinely collected for risk stratification. These complementary measures would provide a comprehensive assessment of risk for both incident heart failure and mortality among individuals with existing heart failure.

TRANSLATIONAL OUTLOOK 1: Future studies, with a causal or quasi-experimental design, should evaluate the impact of public health policies aimed at improving household income, educational attainment, and neighborhood conditions on heart failure incidence.

TRANSLATIONAL OUTLOOK 2: Current and emerging heart failure risk stratification tools should extend beyond neighborhood-level socioeconomic indicators to incorporate individual-level measures, particularly income, which independently and more strongly predicts incident heart failure risk and subsequent mortality.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.